Hall Ticket Number:

MR-21

Code: B415IT5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Discrete Mathematics

(Information Technology)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Construct the truth tables for the logical connectives.	2	1	1	1,2
1. b)	Explain about Normal forms.	2	1	1	1
1. c)	Define Binary Relation and write its properties.	2	1	2	1,2
1. d)	Write about compatibility relation with example.	2	1	2	1,2
1. e)	Write a recursive algorithm to find Fibonacci sequence for a number.	2	2	3	1
1. f)	How many different 3 digit numbers can be formed with 3 four's, 4 two's and 2 three's.	2	2	3	1,2
1. g)	Explain Bayes' theorem.	2	1	4	1
1. h)	Solve recurrence relation $a_n-4a_{n-1}+4a_{n-2}=0$, where $a_0=0$, $a_1=1$.	2	3	4	1,2
1. i)	Define isomorphism and state the rules for isomorphic graphs.	2	1	5	1
1. j)	Explain tree traversals.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that the following statements are logically equivalent without using truth table. $(P\rightarrow R) \wedge (Q\rightarrow R) \Leftrightarrow (P \vee Q)\rightarrow R$	5	2	1	1,2
2. b)	State and explain the rules that can generate a well formed formula.	5	1	1	1
	OR				
2. c)	Prove the following argument, using symbolic logic. All men are fallible. All kings are men. Therefore, all kings are fallible.	5	2	1	1
2. d)	Using the truth table, show that the proposition $P \vee \neg(P \wedge Q)$ is a tautology.	5	2	1	1
3. a)	Suppose that R is the relation on the set of strings that consist of English letters such that a Rb if and only if l(a) = l(b), where l(x) is the length of the string x. Is R an equivalence relation?	5	3	2	1,2
3. b)	Let f& g be functions from R to R defined by f(x) =ax+b and $g(x) = 1-x+x^2$. If (gof) (x)=$9x^2$ -9x+3, determine a,b.	5	3	2	1,2
	OR				
3. c)	Draw the Hasse diagram representing the positive divisors of 36.	5	2	2	1
3. d)	Given X= {1, 2, 3, 4), R= {(x, y)/ x > y}. Draw the graph of R and give its matrix. Also state the properties satisfied by it.	5	3	2	1
4. a)	By using mathematical induction prove that the given equation is true for all positive integers. $1^2 + 2^2 + 3^2 + \ldots + n^2 = (1/6)\{n(n + 1)(2n + 1\}$ for all $n \in N$.	5	2	3	1,2
4. b)	State binomial theorem. Expand $(2a+8b)^{10}$ using binomial theorem.	5	3	3	1,2
	OR				
4. c)	Discuss Pigeon hole principle with a suitable example.	5	1	3	1
4. d)	In how many ways can a committee of 5 ladies and 4 gents be chosen from 9 ladies and 15 gents, if gent A refuses to take part if lady B is on the committee?	5	2	3	1

P.T.O.

5. a)	Find a general expression for a solution to the recurrence relation. $a_n-5a_{n-1}+6a_{n-2}=n(n-1)$ for $n>=2$.	5	3	4	1,2
5. b)	Solve the recurrence relation $a_n+ a_{n-1} - 8 a_{n-2} - 12 a_{n-3} =0$ for $n \geq 3$, given $a_0=1$, $a_1=5$, $a_2=1$ by generating functions.	5	3	4	1,2
	OR				
5. c)	Solve the recurrence relation using substitution method. $a_n= a_{n-1} + 1/ n(n+1)$ where $a_0=1$.	5	3	4	1,2
5. d)	How many 11-permutations can be formed from the word 'TALLAHASSEE'. How many of these permutations begin with T and end with E.	5	2	4	1
6. a)	Show that the complete graph K_5 and complete bipartite graph $K_{3,3}$ are not planar.	5	2	5	1
6. b)	Explain Euler's theorem with an example.	5	1	5	1
	OR				
6. c)	Differentiate between depth first and breadth first graph traversals with an example each.	5	2	5	1
6. d)	Prove that a graph G with at least one edge is 2-chromatic if and only if G has no cycle of odd length.	5	3	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome